

Enhance Digital Government and Economy

Mid Term Exam



Submitted by:

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Case-01:

For 1st Photo:



Result:

```
root@kali: /media/sf_Exam1
File Actions Edit View Help
Device Manufacturer      : Unknown (OPPO)
Device Model             :
Device Attributes        : Reflective, Glossy, Positive, Color
Rendering Intent          : Perceptual
Connection Space Illuminant : 0.9642 1 0.82491
Profile Creator           : Apple Computer Inc.
Profile ID                : 0
Profile Description       : Display P3
Profile Copyright         : Copyright Apple Inc., 2017
Media White Point         : 0.95045 1 1.08905
Red Matrix Column         : 0.51512 0.2412 -0.00105
Green Matrix Column       : 0.29198 0.69225 0.04189
Blue Matrix Column        : 0.1571 0.06657 0.78407
Red Tone Reproduction Curve : (Binary data 32 bytes, use -b option to ext
ract)
Chromatic Adaptation      : 1.04788 0.02292 -0.0502 0.02959 0.99048 -0.
01706 -0.00923 0.01508 0.75168
Blue Tone Reproduction Curve : (Binary data 32 bytes, use -b option to ext
ract)
Green Tone Reproduction Curve : (Binary data 32 bytes, use -b option to ext
ract)
Image Width               : 3000
Image Height              : 4000
Encoding Process           : Baseline DCT, Huffman coding
Bits Per Sample            : 8
Color Components           : 3
Y Cb Cr Sub Sampling       : YCbCr4:2:0 (2 2)
Aperture                   : 1.7
Image Size                 : 3000x4000
Megapixels                 : 12.0
Scale Factor To 35 mm Equivalent: 0.8
Shutter Speed              : 1/50
Create Date                : 2024:09:25 21:16:00.871
Date/Time Original         : 2024:09:25 21:16:00.871+06:00
Modify Date                : 2024:09:25 21:16:00.871
Circle Of Confusion        : 0.035 mm
Field Of View              : 154.9 deg
Focal Length               : 4.7 mm (35 mm equivalent: 4.0 mm)
Hyperfocal Distance        : 0.37 m
Light Value                : 6.9

(root@kali)-[/media/sf_Exam1]
# █
```

1) The photo was taken:

Create Date	: 2024:09:25 21:16:00.871
Date/Time Original	: 2024:09:25 21:16:00.871+06:00
Modify Date	: 2024:09:25 21:16:00.871

2) The device manufacturer:

Device Manufacturer	: Unknown (OPPO)
Device Model	:
Device Attributes	: Reflective, Glossy, Positive, Color
Rendering Intent	: Perceptual

3) Encoding process:

Encoding Process	: Baseline DCT, Huffman coding
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4) Height:

Image Height	: 4000
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5) Original time:

Date/Time Original	: 2024:09:25 21:16:00.871+06:00
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For 2nd Photo:



Result:

```
root@kali: /media/sf_Exam1
File Actions Edit View Help
Profile Version : 4.0.0
Profile Class : Display Device Profile
Color Space Data : RGB
Profile Connection Space : XYZ
Profile Date Time : 2018:06:24 13:22:32
Profile File Signature : acsp
Primary Platform : Apple Computer Inc.
CMM Flags : Not Embedded, Independent
Device Manufacturer : Unknown (OPPO)
Device Model :
Device Attributes : Reflective, Glossy, Positive, Color
Rendering Intent : Perceptual
Connection Space Illuminant : 0.9642 1 0.82491
Profile Creator : Apple Computer Inc.
Profile ID : 0
Profile Description : Display P3
Profile Copyright : Copyright Apple Inc., 2017
Media White Point : 0.95045 1 1.08905
Red Matrix Column : 0.51512 0.2412 -0.00105
Green Matrix Column : 0.29198 0.69225 0.04189
Blue Matrix Column : 0.1571 0.06657 0.78407
Red Tone Reproduction Curve : (Binary data 32 bytes, use -b option to ext
ract)
Chromatic Adaptation : 1.04788 0.02292 -0.0502 0.02959 0.99048 -0.
01706 -0.00923 0.01508 0.75168
Blue Tone Reproduction Curve : (Binary data 32 bytes, use -b option to ext
ract)
Green Tone Reproduction Curve : (Binary data 32 bytes, use -b option to ext
ract)
Image Width : 3000
Image Height : 4000
Encoding Process : Baseline DCT, Huffman coding
Bits Per Sample : 8
Color Components : 3
Y Cb Cr Sub Sampling : YCbCr4:2:0 (2 2)
Aperture : 1.7
Image Size : 3000x4000
Megapixels : 12.0
Scale Factor To 35 mm Equivalent: 0.8
Shutter Speed : 1/17
Create Date : 2024:09:25 22:18:34.529
Date/Time Original : 2024:09:25 22:18:34.529+06:00
Modify Date : 2024:09:25 22:18:34.529
Circle Of Confusion : 0.035 mm
Field Of View : 154.9 deg
Focal Length : 4.7 mm (35 mm equivalent: 4.0 mm)
Hyperfocal Distance : 0.37 m
Light Value : 1.1
(root@kali)-[/media/sf_Exam1]
# █
```

1) The photo was taken:

Date/Time Original	: 2024:09:25 22:18:34.529+06:00
Modify Date	: 2024:09:25 22:18:34.529

2) The device manufacturer:

Device Manufacturer	: Unknown (OPPO)
Device Model	:
Device Attributes	: Reflective, Glossy, Positive, Color

3) Encoding process:

Encoding Process	: Baseline DCT, Huffman coding
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4) Height:

Image Height	: 4000
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5) Original time:

Date/Time Original	: 2024:09:25 22:18:34.529+06:00
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Case-02:

1)

Nmap:

Nmap is a freely available network scanning tool for network discovery and security auditing that is open-source. It enables customers to spot hosts and services on a network by obtaining packets and assessing the results. Nmap could be useful in the detection of open ports, running services, the type of operating system being used and security vulnerabilities on the network devices.

2)

The command to find out own machines IP addresses:

ifconfig

```
(root@kali)-[/media/sf_Exam1]
# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::a00:27ff:fe34:e06a prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:34:e0:6a txqueuelen 1000 (Ethernet)
    RX packets 7 bytes 1266 (1.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 38 bytes 4080 (3.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 4 bytes 240 (240.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 4 bytes 240 (240.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```


3)

The command to find open ports in a target IP

*nmap -sS 192.168.68.216

Target IP=192.168.68.216

```
(root@kali)-[/media/sf_Exam1]
# nmap -sS 192.168.68.216
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-10-05 03:28 EDT
Nmap scan report for 192.168.68.216
Host is up (0.011s latency).
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
139/tcp    open  netbios-ssn
143/tcp    open  imap
443/tcp    open  https
445/tcp    open  microsoft-ds
5001/tcp   open  complex-link
8080/tcp   open  http-proxy
8081/tcp   open  blackice-icecap

Nmap done: 1 IP address (1 host up) scanned in 5.12 seconds
```

4)

The service version on 80 port is STATE SERVICE VERSION

```
(root@kali)-[/media/sf_Exam1]
# nmap -sV 192.168.68.216 -p 80
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-10-05 03:34 EDT
Nmap scan report for 192.168.68.216
Host is up (0.0031s latency).

PORT      STATE SERVICE VERSION
80/tcp    open  http      Apache httpd 2.2.14 ((Ubuntu) mod_mono/2.4.3 PHP/5.3.2-1ubuntu4.30 with Suhosin-Patch proxy_html/3.0.1 mod_python/3.3.1 Python/2.6.5 mod_ssl/2.2.14 OpenSSL ... )

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.72 seconds
```

5)

Command:

*nmap -sV 192.168.68.216 -O

Operating systems running on target system: OS: Linux

```
(root@kali)-[/media/sf_Exam1]
# nmap -sV 192.168.68.216 -O
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-10-05 03:36 EDT
Nmap scan report for 192.168.68.216
Host is up (0.0076s latency).
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 5.3p1 Debian 3ubuntu4 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http         Apache httpd 2.2.14 ((Ubuntu) mod_mono/2.4.3 PHP/5.3.2-1ubuntu4.30 with
          Suhosin-Patch proxy_html/3.0.1 mod_python/3.3.1 Python/2.6.5 mod_ssl/2.2.14 OpenSSL ... )
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
143/tcp   open  imap         Courier Imapd (released 2008)
443/tcp   open  ssl/http     Apache httpd 2.2.14 ((Ubuntu) mod_mono/2.4.3 PHP/5.3.2-1ubuntu4.30 with
          Suhosin-Patch proxy_html/3.0.1 mod_python/3.3.1 Python/2.6.5 mod_ssl/2.2.14 OpenSSL ... )
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
5001/tcp  open  java-object  Java Object Serialization
8080/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
8081/tcp  open  http         Jetty 6.1.25
1 service unrecognized despite returning data. If you know the service/version, please submit the
following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port5001-TCP:V=7.94SVN%I=7%D=10/5%Time=6700EC8E%P=x86_64-pc-linux-gnu%r
SF:(NULL,4,"\\xac\\xed\\0\\x05");
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed p
ort
Device type: bridge|general purpose|switch
Running (JUST GUESSING): Oracle Virtualbox (96%), QEMU (91%), Bay Networks embedded (86%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:qemu:qemu cpe:/h:baynetworks:baystack_450
Aggressive OS guesses: Oracle Virtualbox (96%), QEMU user mode network gateway (91%), Bay Networks
BayStack 450 switch (software version 3.1.0.22) (86%)
No exact OS matches for host (test conditions non-ideal).
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit
/ .
Nmap done: 1 IP address (1 host up) scanned in 21.17 seconds
```